D2.1 Cell and nuclear division

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d2-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 662–669 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

The cell cycle

D2.1.12 Cell proliferation

pp. 662–663

Cells proliferate for **growth** (plant meristems, early animal embryos), **replacement** (skin cells made in the basal layer replace those lost at the surface) and **repair** (healing a wound).

D2.1.13 Phases of the cell cycle

pp. 663–664

The cycle is **interphase** (G1, S and G2), then **mitosis** and **cytokinesis**. DNA is replicated in S. Some cells leave the cycle and stay in G0.

D2.1.14 Growth in interphase

pp. 664–665

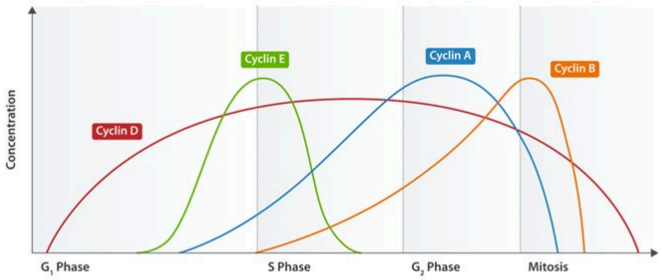
Interphase is metabolically active: the cell makes proteins and membranes, replicates its DNA, and its mitochondria and chloroplasts grow and divide so each daughter cell receives enough.

D2.1.15 Cyclins

pp. 665–666

Cyclins control the cycle

D2.1.15



Cyclin concentrations **rise and fall** in each cycle. Each cyclin binds and activates a **cyclin-dependent kinase** (CDK).

A cyclin must reach a **threshold** for the cell to pass each checkpoint; then it is broken down.

Concentrations of four cyclins through the cell cycle. Public domain.

History

Cyclins were found in sea urchin eggs; the discovery won a Nobel Prize in 2001.

D2.1 Cell cycle and cancer (HL) – Textbook pp. 665–666 10

Concentrations of different **cyclins** rise and fall during the cycle. Each activates a cyclin-dependent kinase, and a **threshold** level of a specific cyclin is needed to pass each **checkpoint**. The cyclin is then broken down.

Cancer

D2.1.16 Mutations in cell cycle genes

pp. 666–667

Proto-oncogenes stimulate division when signalled; a mutation can turn one into an **oncogene** that is always active (for example Ras). **Tumour suppressor genes** stop the cycle if DNA is damaged; mutations that inactivate both copies (for example p53) remove this brake. Either way, cells divide without control.

D2.1.17 Benign and malignant tumours

pp. 667–669

Benign tumours grow slowly and stay in one place; they are not cancer. **Malignant** tumours grow fast, invade nearby tissue and can **metastasize**: cells from the **primary tumour** travel in blood or lymph and form **secondary tumours** elsewhere.

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